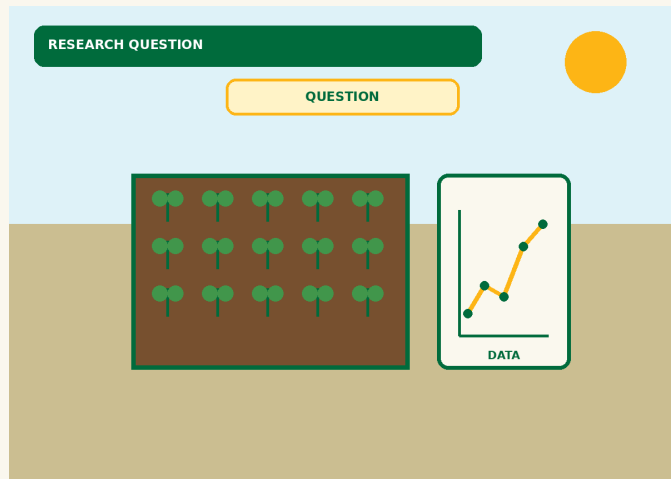




QUESTION

RESEARCH QUESTION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 641 • RESEARCH QUESTION

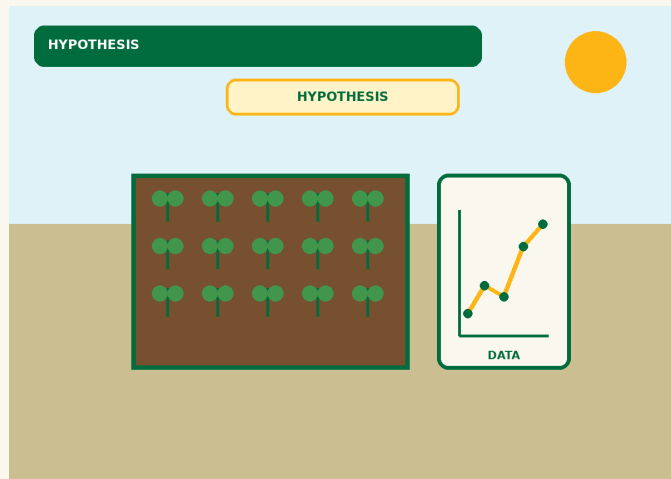
CONCEPT

A research question defines the relationship or difference that an experiment is designed to investigate.

INVESTIGATION TASK

Write a measurable question involving one factor and one response.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



HYPOTHESIS

HYPOTHESIS

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 642 • HYPOTHESIS

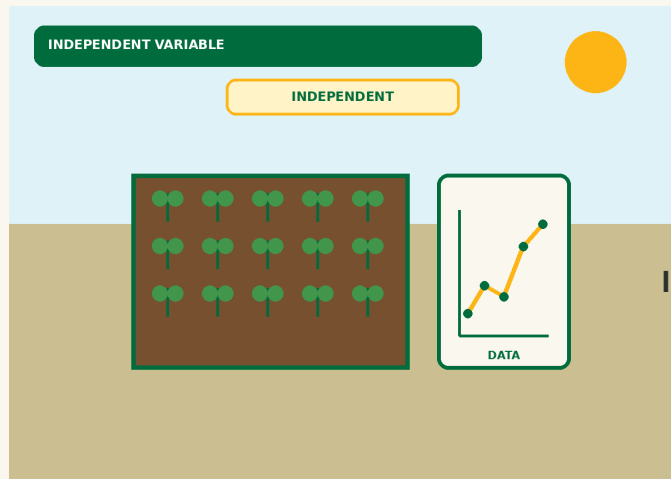
CONCEPT

A hypothesis is a testable prediction about how an independent variable may affect a dependent variable.

INVESTIGATION TASK

Write a directional hypothesis for the trial.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



INDEPENDENT

INDEPENDENT VARIABLE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 643 • INDEPENDENT VARIABLE

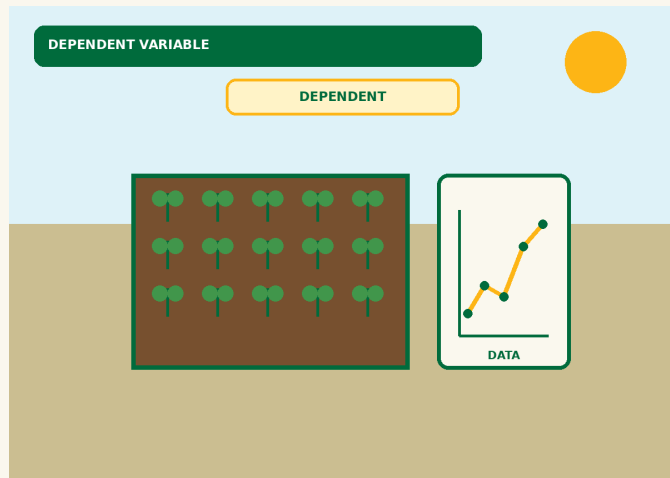
CONCEPT

The independent variable is deliberately manipulated to create experimental treatments.

INVESTIGATION TASK

Specify its levels or treatments.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



DEPENDENT

DEPENDENT VARIABLE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 644 • DEPENDENT VARIABLE

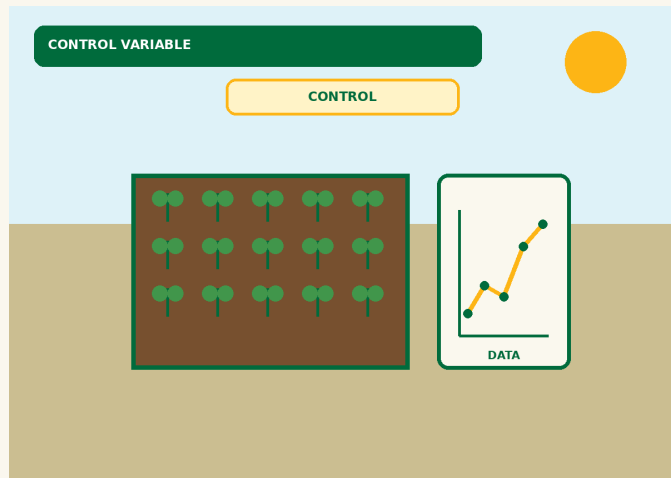
CONCEPT

The dependent variable is the measured response used to evaluate treatment effects.

INVESTIGATION TASK

Define how and when it will be measured.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



CONTROL

CONTROL VARIABLE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 645 • CONTROL VARIABLE

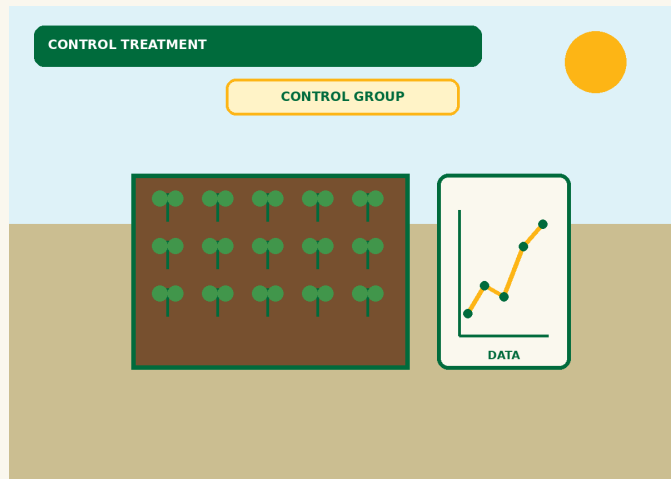
CONCEPT

Control variables are managed consistently to reduce alternative explanations for observed differences.

INVESTIGATION TASK

List three variables to control.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



CONTROL GROUP

CONTROL TREATMENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 646 • CONTROL TREATMENT

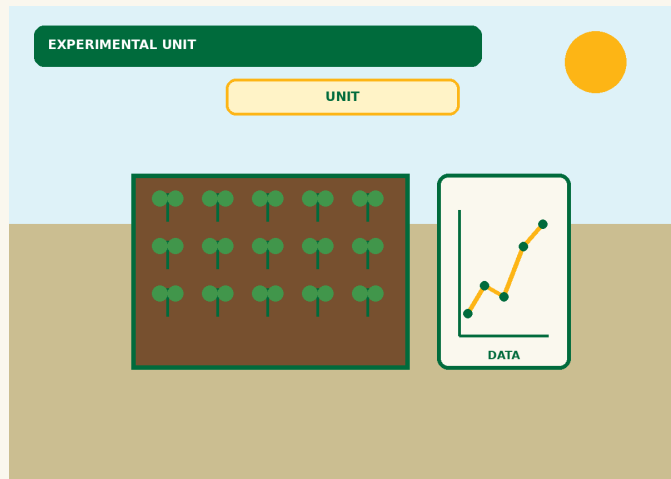
CONCEPT

A control treatment provides a reference condition against which experimental treatments are compared.

INVESTIGATION TASK

Explain why the selected control is appropriate.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



UNIT

EXPERIMENTAL UNIT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 647 • EXPERIMENTAL UNIT

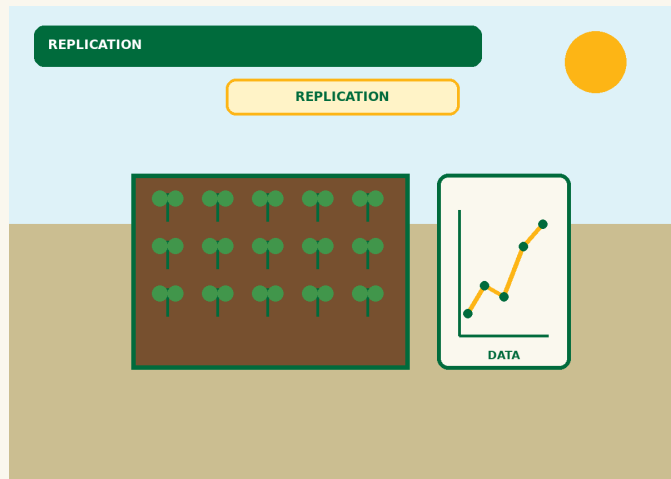
CONCEPT

An experimental unit is the smallest unit independently assigned to a treatment.

INVESTIGATION TASK

Identify the experimental unit in the garden.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



REPLICATION

REPLICATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 648 • REPLICATION

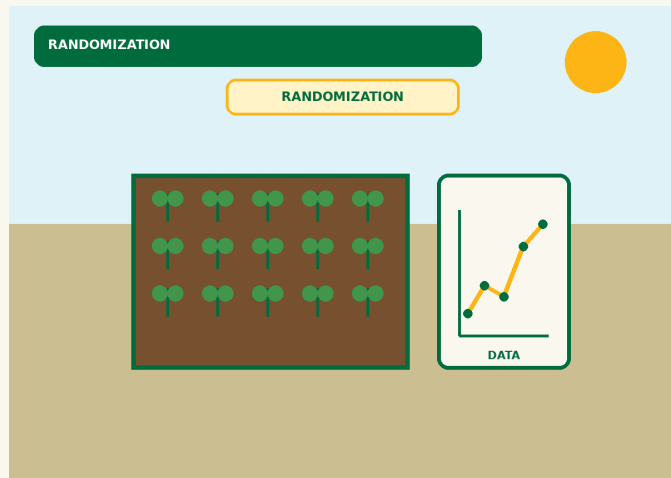
CONCEPT

Replication applies each treatment to multiple independent experimental units to estimate natural variation.

INVESTIGATION TASK

Count replicates and explain their purpose.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RANDOMIZATION

RANDOMIZATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 649 • RANDOMIZATION

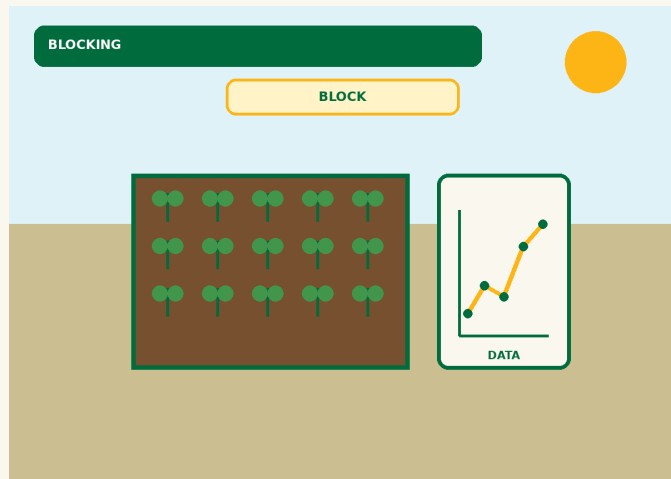
CONCEPT

Randomization assigns treatments by chance to reduce systematic placement bias.

INVESTIGATION TASK

Carry out a teacher-approved random assignment.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



BLOCK

BLOCKING

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 650 • BLOCKING

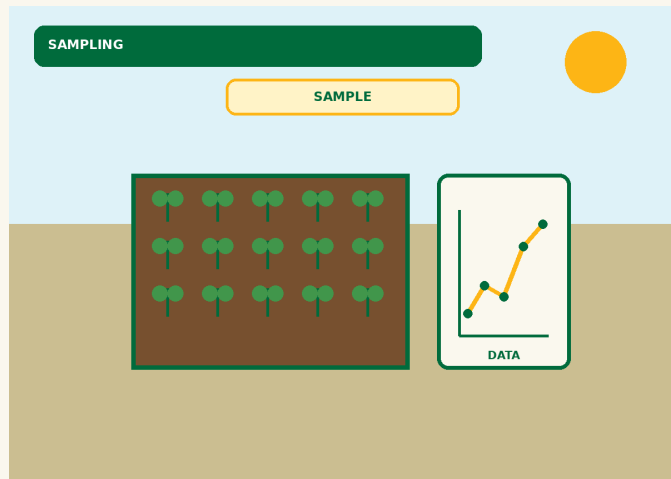
CONCEPT

Blocking groups similar experimental units before randomization to account for known spatial variation.

INVESTIGATION TASK

Design a simple block for a sloping or variable garden.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



SAMPLE

SAMPLING

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 651 • SAMPLING

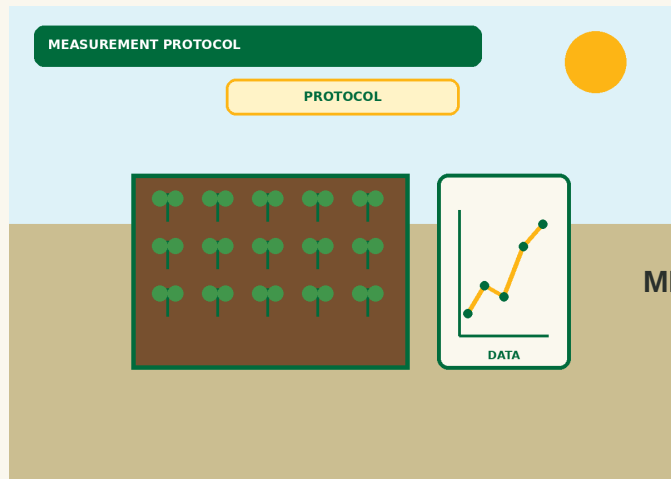
CONCEPT

Sampling selects representative plants, locations, or times for measurement when measuring everything is impractical.

INVESTIGATION TASK

Define a consistent sampling rule.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



PROTOCOL

MEASUREMENT PROTOCOL

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 652 • MEASUREMENT PROTOCOL

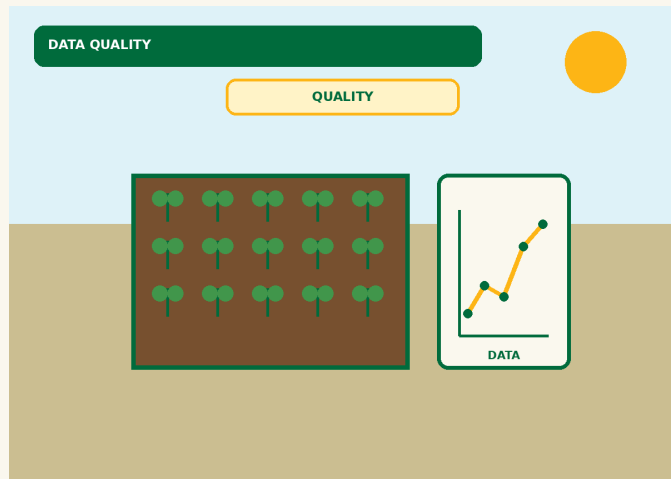
CONCEPT

A measurement protocol specifies the instrument, unit, timing, location, and procedure used to collect data consistently.

INVESTIGATION TASK

Write a protocol for plant height.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



QUALITY

DATA QUALITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 653 • DATA QUALITY

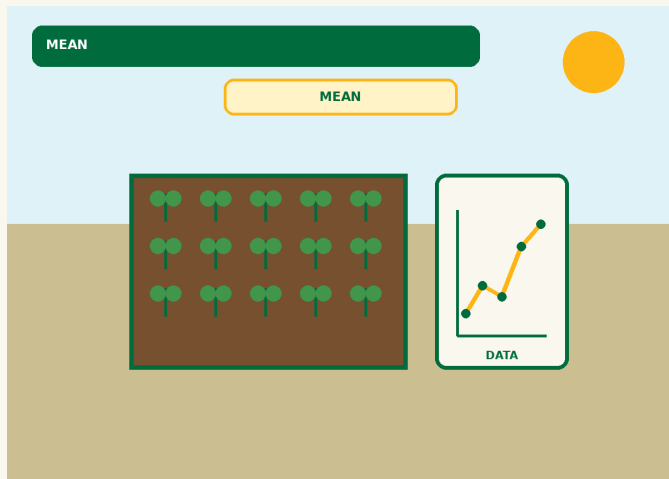
CONCEPT

Data quality depends on consistent methods, correct units, accurate recording, and detection of errors.

INVESTIGATION TASK

Check a sample dataset for inconsistencies.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MEAN

MEAN

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

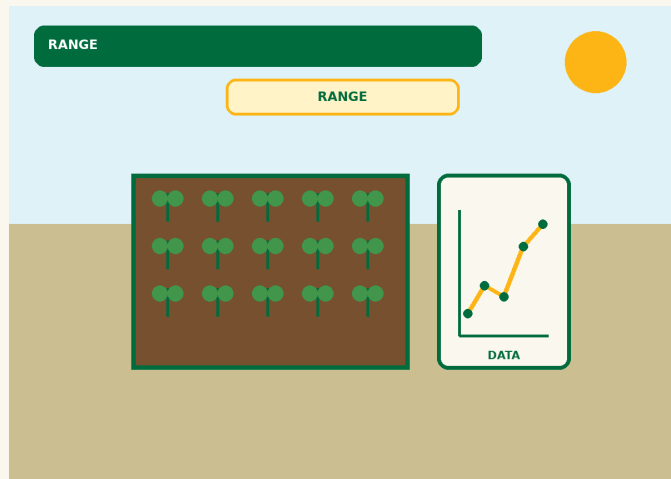
CONCEPT

The arithmetic mean summarizes the central value of repeated numerical observations.

INVESTIGATION TASK

Calculate treatment means.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RANGE

RANGE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 655 • RANGE

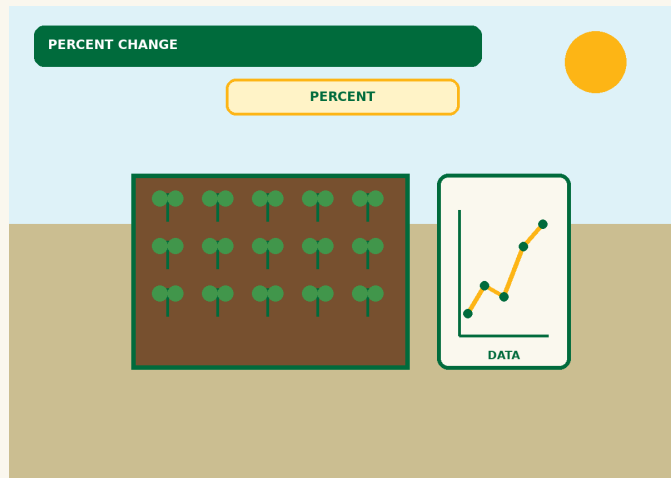
CONCEPT

Range is the difference between the maximum and minimum values and provides a simple measure of spread.

INVESTIGATION TASK

Calculate the range for each treatment.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



PERCENT

PERCENT CHANGE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 656 • PERCENT CHANGE

CONCEPT

Percent change expresses the difference relative to a reference or starting value.

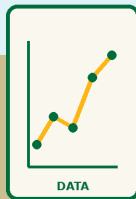
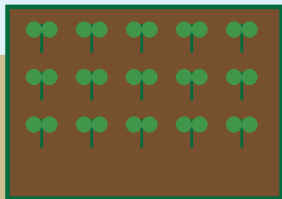
INVESTIGATION TASK

Calculate percent change from baseline.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE

DATA VISUALIZATION

VISUALIZE



VISUALIZE

DATA VISUALIZATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 657 • DATA VISUALIZATION

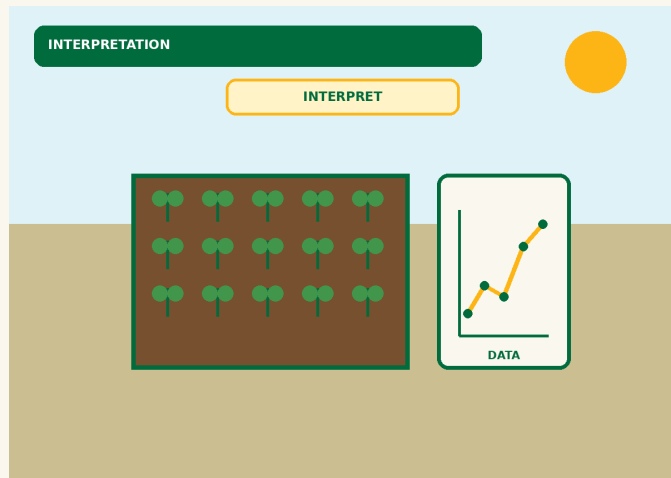
CONCEPT

Graphs and charts reveal patterns, variation, trends, and treatment differences in experimental data.

INVESTIGATION TASK

Choose an appropriate graph for the dataset.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



INTERPRET

INTERPRETATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 658 • INTERPRETATION

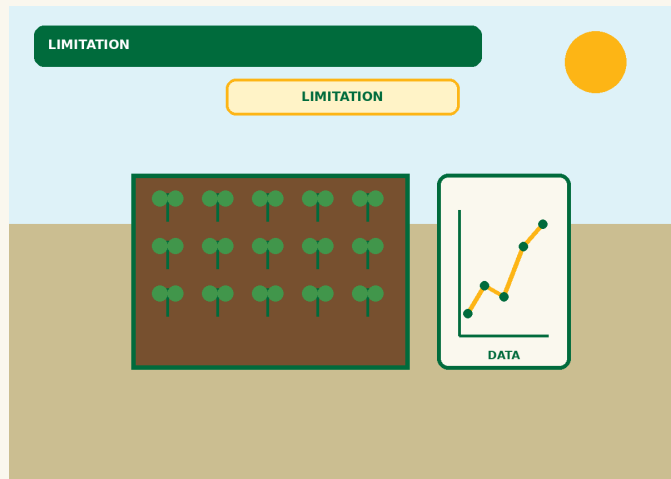
CONCEPT

Interpretation explains what the results indicate while considering variability and experimental design.

INVESTIGATION TASK

Write two evidence-based interpretations.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



LIMITATION

LIMITATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 659 • LIMITATION

CONCEPT

Experimental limitations are factors that reduce confidence, precision, generalizability, or causal interpretation.

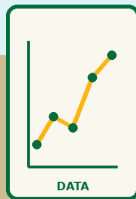
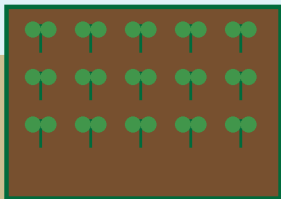
INVESTIGATION TASK

Identify one limitation and its likely effect.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE

EXPERIMENTAL CONCLUSION

CONCLUDE



CONCLUDE

EXPERIMENTAL CONCLUSION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 660 • EXPERIMENTAL CONCLUSION

CONCEPT

A conclusion answers the research question using results, uncertainty, and limitations rather than expectation alone.

INVESTIGATION TASK

State whether the evidence supports the hypothesis.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE